SecuriAgent AI

Complete Technical Architecture & Upgrade Report

Candidate: Kartik Bingi
Role: AI/ML Developer Intern
Target: Kapstone Technological Services
Deadline: October 22, 2026

1. PROJECT VISION & CORE ARCHITECTURE

SecuriAgent AI is an end-to-end intelligent cybersecurity log analyzer and compliance engine designed to bridge machine telemetry and executive governance. Modern Security Operations Centers (SOCs) are overwhelmed by millions of raw, noisy log entries daily. SecuriAgent AI automates anomaly detection, correlates multi-event attack vectors by IP and time windows, maps detected threats to regulatory frameworks, and triggers active sandbox mitigation.

Isolation Forest	FastAPI	Streamlit	ISO / NIST
UNSUPERVISED ML MODEL	ASYNCHRONOUS BACKEND	MULTI-TAB FRONTEND	COMPLIANCE ENGINE

2. ORIGINAL CORE PIPELINE COMPONENTS

Module Name	File Path	Core Responsibility & Mechanics
Synthetic Log Generator	`src_model.py`	Generates realistic security logs simulating normal HTTP traffic alongside attack signatures (brute-force logins, port scans, SQL injection).
Alert Correlator	`src_correlator.py`	Groups raw telemetry by source IP and timestamp windows into cohesive security incidents with dynamic severity scoring (MEDIUM, HIGH, CRITICAL).
Compliance Engine	`src_compliance.py`	Translates technical event signatures into regulatory controls (ISO 27001, NIST CSF) accompanied by actionable remediation advice.
FastAPI Backend	`backend_main.py`	Provides asynchronous REST API endpoints for uploading log files, executing threat detection pipelines, and returning JSON metrics.
Streamlit Dashboard	`app.py`	Interactive visual frontend enabling log upload, real-time metrics view, compliance rendering, and executive report downloads.

3. THREAT DETECTION & SEVERITY MECHANICS

The system evaluates incoming telemetry by analyzing numerical indicators: failure count, payload size (bytes transferred), and HTTP response codes. Anomalous events are categorized into structured incident reports, ensuring critical threat signatures receive prioritized SOC analyst review.

4. UPDATED REQUIREMENTS & UPGRADE STRATEGY

📌 Overview of Directive Updates

Following updated technical assessment instructions from Kalyani, the project was upgraded to replace synthetic UI heuristics with **genuine ML model training and unseen test split evaluation**, alongside an active **sandbox firewall mitigation adapter** that logs state execution.

5. DETAILED TECHNICAL UPGRADE LEDGER

Area	Initial Prototype State	Final Upgraded State
Model Training & Testing	Isolation Forest trained on full dataset without evaluation metrics.	Added 70/30 train-test split, evaluated on unseen test set, serialized precision, recall, & F1-score to `models/evaluation_metrics.json`.
Active Mitigation	Static textual remediation guidelines in tables.	Functional firewall adapter (`src_mitigation.py`) executing active IP blocks to `sandbox/firewall_rules.json`.
Dashboard Architecture	Single-page continuous scrolling view.	3-Tab architecture: Live Analysis, Model Metrics, and Firewall Sandbox Monitor.
Reporting	Static Markdown text string generation.	Dynamic executive security incident report with session metrics and firewall audit trails.

6. MODEL EVALUATION RESULTS (UNSEEN TEST DATASET)

0.963	0.925	0.944
PRECISION SCORE	RECALL SCORE	F1-SCORE

7. COMPLIANCE & MITIGATION MAPPING TABLE

Detected Event	NIST CSF Reference	ISO 27001 Reference	Mitigation / Sandbox Action
LOGIN_FAILURE	PR.AC-1 (Access Control)	A.9.4.2 (Secure Log-on)	Enforce account lockouts after 5 failed attempts.
PORT_SCAN	DE.CM-1 (Network Monitoring)	A.13.1.1 (Network Controls)	Execute sandbox firewall drop rule (`DROP_ALL_TRAFFIC`).
SQL_INJECTION	PR.DS-5 (Data Protection)	A.14.1.2 (Application Security)	Implement parameterized queries and WAF filtering.

8. ACTIVE SANDBOX FIREWALL EXECUTION MECHANICS

The newly developed mitigation module (`src_mitigation.py`) allows security operators to select high-risk source IPs identified by the anomaly model and execute a simulated firewall block rule. The adapter checks for existing block entries to prevent duplication and appends a structured record containing the target IP, enforcement action (`DROP_ALL_TRAFFIC`), policy reason, and execution timestamp into `sandbox/firewall_rules.json` .

9. FINAL ASSESSMENT COMPLIANCE MATRIX

Assessment Directive	Verification Evidence in Final Project	Status
1. Genuine Model Training	Unsupervised Isolation Forest model trained, saved to `models/isolation_forest.pkl` , and evaluated on unseen split.	VERIFIED
2. Threat Detection Pipeline	Log ingestion pipeline running real inference via standard scaling and model predictions.	VERIFIED
3. Compliance Mapping	ISO 27001 and NIST CSF control mappings dynamically generated for flagged anomalies.	VERIFIED
4. Sandbox Mitigation Action	Functional IP drop rule execution storing persistent state in `sandbox/firewall_rules.json` .	VERIFIED
5. API & Executive Reporting	FastAPI backend operational; downloadable Markdown security reports available in Streamlit dashboard.	VERIFIED

10. CONCLUSION & FUTURE SCOPE

Conclusion: SecuriAgent AI successfully delivers a working, end-to-end prototype combining genuine machine learning anomaly detection, alert correlation, regulatory compliance mapping, and active sandbox firewall mitigation. All deliverables, evaluation artifacts, backend API endpoints, and frontend dashboards are fully functional and verified.

Future Scope: Future enterprise iterations can expand into real-time SIEM integrations (Elasticsearch/Splunk connectors), deep sequential neural network threat models (LSTM/Transformers), and automated SOAR webhook triggers for live network infrastructure.